

IG Agent v25 - Complete Final Specification

FINAL v8 | June 2026 | CONFIDENTIAL

Field | Value

Application version | **25.5.0**

Spec version | **v8** (supersedes v1-v7)

Foundation | v24-proven (25 May 2026)

Date | June 2026

Status | **SHIPPED** - main @ 3ae5aa1

North-star doc | This file + `IG\_Agent\_v25\_COMPLETE\_SPEC\_v8.pdf`

Purpose - Read This First

This is the complete, final specification for IG Agent v25 as implemented and verified through v25.5.0 (June 2026). It supersedes all previous spec versions (v1-v7) and reflects code on main, not aspirational design.

v8 update (June 2026): Incorporates all development since spec v7 (May 2026):

- Multi-market live operation (Japan 225, Wall Street, Spot Gold, US Tech 100)
- ML confidence blend operational (`USE\_ML\_SIGNAL=true`)
- Corrected IG contract point values (`ig\_point\_value\_gbp`)
- Lifecycle hardening: graceful shutdown, post-exit verification, manual-stop flag, Safe to Leave
- Dashboard audit (P0-P2): trades dedupe, points CAUTION band, REST /3, changelog splash
- Strategy Help modal (in-dashboard operator reference)
- Startup splash with phase checklist; cold-start bars reduced to 2
- Gap-open block expiry at 12 bars; environment fitness pass $\geq 55\%$
- Seven-gate trading loop (documented accurately - v7 "13 gates" mixed pipeline stages with entry gates)
- 500+ pytest suite; cold-start E2E audit script

Operator quick reference: Dashboard -> ? Strategy help (mirrors Sections 4-8 below).

1. What v25 Is - The Step Change

v25 is three step changes in one:

Step | Change

****1. GUI**** | Tkinter -> React + FastAPI web dashboard (`localhost:8080`)

****2. AI/ML**** | Points reinforcement + XGBoost blend + nightly replay pipeline

****3. Engine**** | v24 operational fixes built in: async orders, position sync, REST budget, deal reconciliation

v24 -> v25 - Proven vs Shipped

Dimension | v24 Proven | v25 Shipped (v8)

GUI | Tkinter, beach-ball freezes | React/Vite dashboard, WebSocket ticks

Signal gate | Confidence ≥ 92 only | Rules score + ML blend + fitness + points (effective floor 80% in HEALTHY)

Exit management | Fixed stop/target | ATR-scaled breakeven/trail + optional limit extension

Learning | None | Points engine + setup memory + ML training store

Trade integrity | Phantom trades (fixed) | IG-confirmed closes, BREAKEVEN/PENDING labels, dedupe
Session mgmt | Manual restart | Auto refresh, gap protection, session-end flatten
Order execution | Sync blocked loop | Async execution engine
Position sync | Stopped on watchdog kill | Independent sync thread
Watchdog | 60s order timeout | 300s startup grace, manual-stop respect, launchd
Markets | Primarily Japan 225 | 4 enabled CFD indices/commodities
Monitoring | Tab-based Tkinter | 5-tab web dashboard + Safe to Leave + Strategy Help

2. Lessons Learned from v24 - Built Into v25

All v24 production incidents (25 May 2026) have explicit mitigations:

- **REST budget deadlock** -> 3 calls/min hard cap, order in-flight 30s timeout
 - **Quote staleness** -> Lightstreamer hub only; 45s tick max age
 - **Duplicate agent processes** -> Instance lock + startup orphan cleanup
 - **Deal ID mismatch** -> Dual ID matching + transaction sync singleton
 - **Watchdog kill during order** -> Extended grace + async execution
 - **Wrong P&L currency** -> GBP-normalised learning store; phantom dry-run excluded from daily P&L
 - **Spread saturation at open** -> 2.5x normal spread cap (not fixed pts only)
 - **IG point value wrong** -> Per-instrument `ig\_point\_value\_gbp` (Nasdaq 14.94, Dow 7.87, etc.)
-

3. Three-Speed Learning System

Speed | Mechanism | Status

Live | Every closed trade -> points engine + learning DB + ML training hooks | **Operational**

Shadow | Every `SignalEngine.evaluate()` -> `shadow\_log.jsonl` | **Operational**

Replay | Nightly scheduler -> historic bars at speed -> dataset + optional retrain | **Operational** (optional overnight)

3.1 Nightly Replay Cycle

- Scheduler: `src/system/replay\_daily\_scheduler.py` / `scripts/` cron examples
- Blocked during live window where configured (22:30-07:00 BST typical)
- Output: replay analysis, training dataset builder, optional XGBoost retrain

3.2 Label Weighting

- Live closes weighted highest; replay rows tagged in dataset
 - ML training uses **fired signals only** (`fired=true` rows)
 - BREAKEVEN labels excluded from binary training target
-

4. Points-Based Reinforcement System

Source: src/trading/points_engine.py

Persisted: src/data/state/points_state.json

4.1 Scoring Formula

After 5+ confirmed closes (rolling 20), scoring is confidence-band weighted:

Result | Band | Points formula

WIN | high ($\geq 92\%$) | $+3 \times (\text{pnl} / \text{avg_win})$

WIN | standard (85-91%) | $+2 \times (\text{pnl} / \text{avg_win})$

WIN | marginal (80-84%) | $+1$ flat

WIN | low ($< 80\%$) | 0

LOSS | high | $-4 \times (\text{loss} / \text{avg_loss})$

LOSS | standard | $-2 \times (\text{loss} / \text{avg_loss})$

LOSS | marginal | -1 flat

BREAKEVEN | any | 0

Before 5 confirmed trades: flat $+1 / -1 / 0$.

4.2 Staged Backing-Off and Recovery

Nominal state (from cumulative points):

Cumulative | Nominal state

$> +6$ | HEALTHY

≥ 5 to $+6$ | CAUTION

≥ 30 to ≥ 5 | WARNING

$< \geq 30$ | STOP (latched)

Effective state may improve with recovery wins (3 $\rightarrow$ one notch better; 5 $\rightarrow$ HEALTHY boost).

State | Entry confidence bar | Size multiplier (see §8)

HEALTHY | $\geq 80\%$ (floor rises with bootstrap wins) | Tiered 1x-4x by cumulative + band

CAUTION | $\geq 80\%$ | $0.5 \times$ flat (v8: was split 0.25/0.5 in v7-era code)

WARNING | $\geq 92\%$ only | 0.25x on high band only

STOP | No entries | 0x

Session guards:

- **6 consecutive losses** $\rightarrow$ skip next **1** actionable signal (session pause)
- **$> \text{GBP}2000$ realised loss in 60 min** $\rightarrow$ forced WARNING for 30 min
- **Daily GBP halt:** `max_daily_loss_gbp = GBP500` (points_state gate; day-stop via session score disabled)

Confidence bands: high $\geq 92\%$, standard $\geq 85\%$, marginal $\geq 80\%$.

5. Exit Management - ATR Trailing Stops

Source: src/trading/trade_manager.py

Mechanism | Implementation (config-driven)

Initial stop | Entry $\pm$ adaptive risk points (ATR-based when enabled)

Initial target | Entry $\pm$ limit (risk $\times$ reward multiple, ATR-capped)

Breakeven | `breakeven_trigger_points` (30) **or** `breakeven_trigger_atr_multiple` $\times$ entry ATR (0.4 for Nasdaq tuning)

Trailing | `adaptive_trailing_trigger_points` (50) **or** `trail_trigger_atr_multiple` $\times$ ATR (0.75)

Trail distance | `adaptive\_trailing\_distance\_points` (25) or ATR band from entry meta
Limit extension | Optional (`limit\_extension\_enabled`): push target by ATR steps, max 3 extensions
Max age | `max\_position\_age\_minutes` = 480 (8 h)
Session flatten | `auto\_flatten\_on\_session\_end` - close all T?N min before session end
Broker stops | IG `PUT /positions/otc/{deal\_id}` when `broker\_stop\_management` enabled

Iron rule: Trail and breakeven only move stop in the profit direction.

Partial close: Config stub present (partial_close_enabled: false) - not yet implemented in v25.5.0.

6. Session Management

Source: src/trading/session_manager.py, src/signals/indicators.py

Requirement | Implementation

Auto session refresh | Closed->open transition starts fresh session state

Cold start block | `**2 bars**` (`COLD\_START\_BARS=2`) - reduced from v7's documented 6; OHLC pre-seed warms indicators

Gap protection | Gap >1x ATR blocked until `**12 bars**` (`GAP\_CLEAR\_BARS`) or 60 min wall-clock

Session whitelist | Per-instrument `trading\_session\_whitelist` enforced in `session\_open` gate

BST session names | `asia\_early` 00-07, `london\_morning` 07-12, `london\_us\_overlap` 12-16, `us\_afternoon` 16-22, `late` 22-00

Japan 225 maintenance | Hub maintenance flag + `japan225\_strategy\_paused`

Entry near close | Blocked when session ends within configured minutes

REST at open | Staggered OHLC bootstrap (22s between REST fetches)

7. Environment Fitness System

Source: src/trading/environment_scorer.py

Gate threshold: GATE_PASS_MIN = 55 (v7 documented 40 - updated in v8)

Four factors (ATR, trend, session, spread) plus client sentiment adjustment. Cold-start and gap-open may cap displayed score. Sentiment crowded long/short applies fitness penalty (Block E).

Dashboard header shows Fit score; LIVE tab shows factor breakdown.

8. Entry Gates - Seven-Gate Decision Flow

Source: src/api/snapshot.py -> GATE_NAMES; evaluated in src/trading/trading_loop.py

v7 listed 13 "gates" mixing entry checks with post-entry monitoring. v8 documents the actual seven pre-trade gates displayed on the dashboard LIVE tab:

| Gate | Pass condition | Fail action

1 | `session\_open` | Market open, not maintenance, in session whitelist, not near session end | WAIT - log detail

2 | `cold\_start\_gap` | Not in cold start (2 bars); no uncleared gap (>1x ATR) | WAIT

3 | `environment\_fitness` | Score >= 55% | WAIT

4 | `points\_state` | Not STOP; not session pause; daily loss < GBP500 | WAIT

5 | `risk\_validation` | Spread <= 2.5x normal; position slots; risk <= cap (size clipped if possible) | WAIT

6 | `signal\_confidence` | BUY/SELL and confidence >= effective threshold; includes ****ML blend**** | WAIT

7 | `execution` | auto_trade, adaptive checks, correlation guard, live arming, no in-flight order | WAIT / submit async

Sub-checks inside gates (not separate dashboard gates):

- Market suspension (`risk\_validation`)
- Correlation guard: max ****15**** new entries per direction per calendar day (`execution`)
- Adaptive engine blocks: bad setup, low confidence, wide spread (`execution`)
- Circuit breaker: 5 losses -> 60 min pause, half size on resume

8.1 Signal Confidence & ML Blend

Rules engine (src/signals/signal_engine.py): EMA 9/21, RSI 14, ATR 14 on closed 5m bar; 15m trend filter.

Default global signal_threshold / confidence_floor = 80. Per-instrument overrides (e.g. Japan 225: 70).

ML blend (inside gate 6, not separate gate):

- `USE\_ML\_SIGNAL=true`; XGBoost model in `src/data/ml\_model/`
- Features: `adjusted\_score`, `raw\_score`, `rsi`, `atr\_ratio` (ATR ÷ stop distance)
- Blend when |prob ? 0.5| >= ****0.15****: `conf = rulesx0.6 + probx100x0.4`, capped 0-100
- Near-50% prob -> rules only (no veto)
- Rolling last-20 decisions in dashboard INTELLIGENCE / LIVE ML log

8.2 Risk Validation & Position Sizing

Base size from instrument trade_size, multiplied by points_engine.get_size_multiplier().

Market | Base size | Stop (pts) | Risk cap | GBP/pt | Min conf.

Japan 225 | 0.4 | 45 | GBP100 | 5.13 | 70%

Wall Street | 0.3 | 80 | GBP150 | 7.87 | 70%

Spot Gold | 6.0 | 10 | GBP200 | 0.79 | 80%

US Tech 100 | 0.25 | 100 | GBP150 | 14.94 | 75%

Clamps: adaptive_min_trade_size 0.01 - adaptive_max_trade_size 50; IG min deal size; clip to risk_cap_gbp.

Position limits: max_open_positions = 15 total; max_positions_per_epic = 2 base.

Dynamic ladderling (HEALTHY only): all epic positions profitable -> +1 slot; oldest >=20 min -> +2 slots max.

Cooldown: 180 s between entries on same epic.

9. Reliability Controls (Blocks A-F) - Status

Block | Area | v8 status

A | REST budget, margin preflight, suspension, spread median | ****Shipped****

B | Replay pipeline, scheduler | ****Shipped****

C | ML hooks, autopsy, dataset, shadow log, scorer | ****Shipped**** (ML on)

D | Intelligence tab APIs | ****Shipped****

E | Client sentiment | ****Shipped****

F | Session/points persistence | ****Shipped****

10. Historical Data Ingestion

- Cache: `src/data/ohlc\_cache/{market}\_5m.jsonl`
 - Status: `src/data/state/ohlc\_pull\_status.json`
 - Yahoo seeder fallback for missing cache (incl. Nasdaq)
 - Parallel warm load + staggered REST fetch at startup
-

11. ML Pipeline

Component | Path | Status

Training store | `ml\_training\_store.jsonl` | Active

Shadow log | `shadow\_log.jsonl` | Active

Dataset builder | `scripts/` / training pipeline | Active

Model | `src/data/ml\_model/model.pkl` | Loaded when present

Scorer | `src/trading/ml\_scorer.py` | Blend at gate 6

Retrain | Replay scheduler / manual | Optional

12. Web Dashboard

12.1 Technology Stack

Layer | Technology

Backend | FastAPI on port 8080

Real-time | WebSocket `/ws` + 5s REST poll fallback

Frontend | React 18 + Vite + Tailwind

Build | `dashboard/dist/` served by agent

12.2 Five Panels

Tab | Contents (v25.5.0)

****LIVE**** | Gates (7), signal confidence, ML log, Why No Trade, market pills

****TRADES**** | Active positions, close buttons, closed trades (deduped), BREAKEVEN/PENDING

****POINTS**** | Cumulative/session/trade scores, state bands, 0.5x CAUTION tooltip

****INTELLIGENCE**** | Shadow stats, learning progress, replay summary

****SYSTEM**** | REST /3, uptime, position sync, stop/restart, emergency controls

12.3 Header Barometers

Bid/Offer, stream status, Today P&L, Win (last 20), Fit, Pos count, points pills, sentiment badge, Strategy help, Safe to Leave, Stop Agent.

12.4 Splash & Lifecycle UX (v8 new)

Feature | Behaviour

Startup splash | Phase checklist on cold start (`StartupSplash.jsx`)

Changelog splash | Once per version (`localStorage` + `version.json`)
Stop Agent | `POST /api/shutdown` -> cleanup -> post-exit verifier on :8081
Manual stop flag | `src/data/state/manual\_stop.json` - watchdog won't auto-restart
Safe to Leave | 13 overnight trust checks (`scripts/safe\_to\_leave.py`)
Strategy Help | In-dashboard operator guide (`strategyHelp.js`)

12.5 Key API Endpoints

Endpoint | Purpose

`GET /state`, `WS /ws` | Dashboard snapshot
`GET /api/startup/status` | Startup phases
`GET /api/splash` | Version changelog
`POST /api/shutdown` | Graceful stop
`GET /api/shutdown/verify-status` | Post-stop verification
`POST /api/safe-to-leave` | Overnight checklist
`GET /api/replay/summary` | Replay results
`GET /api/shadow/today` | Shadow signal stats
`GET /api/learning/status` | ML / DB progress

13. Client Sentiment (Block E)

On session open: IG client sentiment cached per epic.
crowded_long -> BUY fitness penalty; crowded_short -> SELL penalty.
Dashboard header badge when active.

14. Session & Points Persistence (Block F)

Atomic JSON writes via state_manager. Points, correlation guard, manual stop, shutdown verify snapshot, rate-limit state - all survive restart.

15. Complete System Architecture

15.1 Core Components

Component | Responsibility

`main.py` | FastAPI + agent bootstrap
`MarketOrchestrator` | One thread per epic
`TradingLoop` | 5s tick, seven gates, snapshot publish
`SignalEngine` | Rule-based scoring
`MLScorer` | XGBoost probability
`PointsEngine` | Cumulative scoring + sizing
`EnvironmentScorer` | Fitness factors
`AdaptiveEngine` | ATR stops, setup multipliers

`ExecutionEngine` | Async IG orders
`TradeManager` | Breakeven, trail, exits
`IgPositionSync` | REST position reconciliation
`Lightstreamer hub` | Live quotes
`RestApiBudget` | 3 calls/min
`Watchdog` | Auto-restart unless manual stop

15.2 Post-Entry Monitoring (not entry gates)

Stage | Responsibility
Trail monitoring | `TradeManager.update\_from\_quote` each tick
Position sync | 20-60s interval
Session flatten | T?N min before close
Health monitor | Trading loop heartbeat
Drawdown | Daily GBP500 cap via points gate

16. Data Files & Operator Artefacts

Path | Purpose
`config/config\_v25.json` | Primary configuration
`src/data/learning\_db.sqlite3` | Trades, learning stats
`src/data/state/points\_state.json` | Points cumulative state
`src/data/state/dashboard\_snapshot.json` | Last published snapshot
`src/data/state/manual\_stop.json` | Deliberate stop flag
`src/data/.ig\_agent\_v25.lock` | Single-instance lock
`src/data/logs/engine.log` | Primary log
`src/data/logs/rate\_limit\_state.json` | REST backoff
`scripts/confirm\_stopped.py` | Verify clean shutdown
`scripts/live\_e2e\_audit.py` | Cold-start API audit
`scripts/pre\_flight\_check.py` | Pre-live checklist

17. Configuration

Loader: config_v25.json -> config_v24.json -> legacy fallback.

Critical globals (v25.5.0):

Key | Value | Notes
`signal\_threshold` | 80 | Pre-filter + gate floor
`confidence\_floor` | 80 | Bootstrap rises +1/win
`USE\_ML\_SIGNAL` | true | Blend at gate 6
`max\_daily\_loss\_gbp` | 500 | Hard daily halt
`max\_open\_positions` | 15 | Portfolio cap
`max\_positions\_per\_epic` | 2 | + dynamic laddering
`cooldown\_seconds` | 180 | Per-epic entry spacing
`rsi\_buy\_max` | 85 | Hard RSI block

`rsi\_sell\_min` | 15 | Hard RSI block

`auto\_flatten\_on\_session\_end` | true | Session close flatten

Per-instrument overrides in instruments.{key}.*.

18. Operations

18.1 Launch

- Desktop icon -> `launcher/IG Agent v25.app` -> `launch.sh`
- Fast path (~1s): agent already healthy on :8080 -> open browser only
- Cold path (~30-45s): full startup + splash

18.2 Stop

1. Dashboard ****Stop Agent**** or `POST /api/shutdown`
2. Sets manual-stop flag; spawns `shutdown\_verify\_server.py`
3. Verify: `PYTHONPATH=src python3 scripts/confirm\_stopped.py`

18.3 Watchdog & launchd

- `scripts/watchdog.sh` - 30s check; respects `manual\_stop\_active`
- `scripts/install\_launchd.sh` - agent + caffeinate + watchdog plists
- Stale lock after crash: `rm -f src/data/.ig\_agent\_v25.lock`

18.4 Pre-Live Checklist

```
PYTHONPATH=src python3 scripts/pre_flight_check.py --live
PYTHONPATH=src python3 scripts/e2e_platform_validation.py
POST /api/safe-to-leave # expect 13/13
```

18.5 Tests

```
PYTHONPATH=src python3 -m pytest tests/ -x -q
```

19. Multi-Instrument Architecture

19.1 Enabled Instruments (v25.5.0)

Key | Epic | Priority

`japan\_225` | IX.D.NIKKEI.IFM.IP | default

`wall\_street` | IX.D.DOW.IFM.IP | 70

`gold` | CS.D.CFP.GOLD.CFP.IP | 100

`nasdaq\_100` | IX.D.NASDAQ.IFM.IP | 65

Disabled in config but scaffolded: EUR/USD, GBP/USD, US Oil, Germany 40.

19.2 Activation

Set enabled: true in instrument block; restart agent; OHLC bootstrap + orchestrator thread starts automatically.

20. Delivery Status - Eight-Week Plan

Phase | Target | v8 status

Core engine + v24 fixes | Week 1-2 | ****Done****

Web dashboard | Week 3-4 | ****Done**** (5 tabs)

Points + learning | Week 4-5 | ****Done****

ML pipeline | Week 5-6 | ****Done**** (blend on)

Multi-market + hardening | Week 6-8 | ****Done**** (v25.5.0)

Live gate (16 checks) | Pre-live | ****Scripts ready**** - run Safe to Leave

21. Live Trading Gate - Operator Checklist

Before unattended operation:

1. Agent HEALTHY on `/api/health`
 2. Safe to Leave 13/13
 3. Watchdog active OR deliberate manual-stop understood
 4. `max\_daily\_loss\_gbp` appropriate for account
 5. Demo/LIVE mode verified in config
 6. Dashboard Strategy Help reviewed for current thresholds
-

22. Component Safety and Performance

- ****Safe defaults:**** Gate/scorer exceptions -> WAIT, not trade
 - ****ML timeout:**** 0.5s subprocess cap; falls back to rules
 - ****Quote freshness:**** 45s max tick age
 - ****REST:**** 3/min atomic budget
 - ****Performance:**** One thread per market; snapshot publish per tick
-

23. Operational Safety

Control | Implementation

Emergency stop | Dashboard + `scripts/emergency\_stop.sh`

Flatten all | `POST /api/flatten/all`

Multi-instance | Instance lock + orphan kill on startup

Manual stop | Prevents watchdog restart until icon launch

Drawdown | GBP500 daily + points STOP latch

24. Absolute Rules - Non-Negotiable

1. **Never** trade without IG-confirmed position sync on live
 2. **Never** run two agents (lock file)
 3. **Never** commit credentials to disk
 4. **Never** bypass points STOP without explicit operator reset
 5. **Never** edit `points\_state.json` or SQLite WAL while agent running
 6. Dashboard is **read-only** for trading state (controls except authorised POST endpoints)
 7. Rebuild `dashboard/dist` after any `dashboard/src` change
-

25. Current Codebase State (June 2026)

Item | Status

Branch | `main` @ 3ae5aa1

App version | 25.5.0

Spec | v8 (this document)

PR | #4 merged - lifecycle + dashboard audit

Tests | 500+ pytest passing

E2E audit | `live\_e2e\_audit.py` - 19 API checks

Markets live | Japan 225, Wall St, Gold, Nasdaq

ML | Trained model blend active

Tkinter | Removed

Dashboard | React dist served on :8080

26. v7 -> v8 Change Log (Spec Corrections)

Topic | v7 doc | v8 actual

Gate count | 13 entry+monitor gates | **7** pre-trade gates on LIVE tab

Fitness pass | $\geq 40\%$ | **$\geq 55\%$**

Production confidence | 92% default | **80%** floor (WARNING 92%)

ML gate | Separate gate 9 | **Blend inside gate 6**

CAUTION size | 0.75x | **0.5x** flat $\geq 80\%$

HEALTHY threshold | $> +10$ pts | **$> +6$ pts**

STOP threshold | $< ?30$ (same) | Same; latched

Cold start | 6 bars | **2 bars**

Gap clear | vague | **12 bars** / 60 min

Spread cap | 1.5x implied | **2.5x** normal

GBP/pt | 1.0 implied | **Per-instrument** (Nasdaq 14.94)

Session loss pause | 3 skips after 3 losses | **1 skip after 6 losses**

Day stop | session score $< ?5$ | **Disabled** - use GBP500 daily

Partial close | Specified | **Config stub only**

Dashboard tabs | 4 + intel | **5 tabs** + Strategy Help

Shutdown | Basic | **Verify server + manual stop**

IG Agent v25 - Complete Final Specification v8 - Confidential

Web Dashboard | Points Reinforcement | ML Blend | Multi-Market | Lifecycle Hardening | June 2026